



CONGA: Continual Neural Gated Architecture for Long-History Sequential Recommendation

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Outline

1. Motivation and research gaps
2. The CONGA architecture
3. Experimental protocol
4. Results (RQ1 – RQ4)
5. Discussion, limitations, conclusion

Sequential Recommendation: A Static Architecture

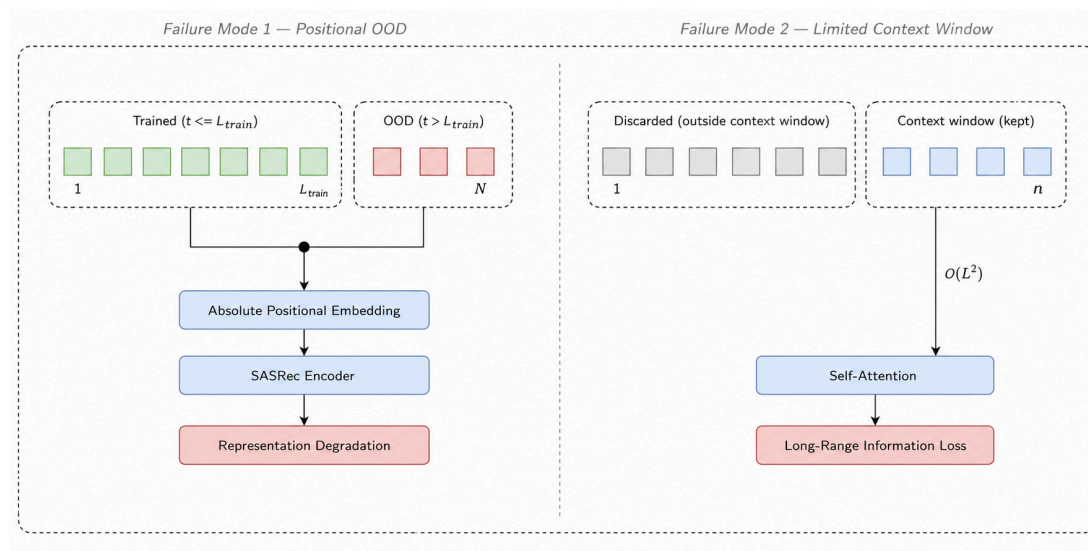
Sequential recommendation predicts a user's next interaction from their chronologically ordered history – jointly modeling temporal dynamics and evolving preferences.

Problem statement

Given interaction sequence $S^u = [s_1^u, \dots, s_{n_u}^u]$ of user u , learn a scoring function $f_\theta : \mathcal{I}^* \rightarrow \mathbb{R}^{|\mathcal{I}|}$ that ranks the next item $s_{n_u+1}^u$ highest among the **full catalog**.

SASRec (2018) established causal self-attention as the dominant paradigm. Yet its core design has stayed largely static – even as RoPE, gated activations, and external memory drove substantial gains in language modeling.

Two Structural Failure Modes of SASRec



(1) **Positional OOD degradation** – absolute position embeddings are learned per index; any $t > L_{train}$ is out-of-distribution by construction (ML-1M histories reach 300–600 interactions).

(2) **Hard context window** – $O(L^2)$ attention restricts practical windows to 200–300 items, and a single-stream residual forces one d -dim space to hold both short- and long-term preference.

Research Questions and Contributions

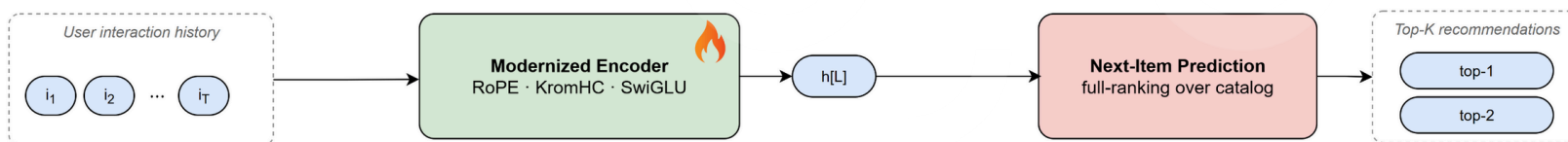
- RQ1** Does CONGA outperform SOTA baselines under full-ranking evaluation?
- RQ2** How much does each architectural component contribute, and how does that vary with data density?
- RQ3** Does CONGA preserve catalog diversity and avoid popularity bias?
- RQ4** Where does TITANS memory's benefit concentrate across users?

Three contributions

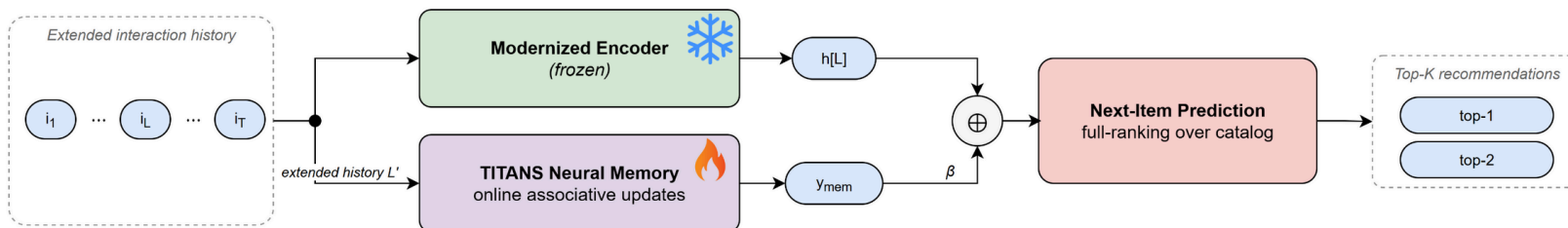
- C1** Diagnose + fix positional OOD via RoPE; isolate KromHC's gain to balanced gradient flow, not extra parameters.
- C2** First proof-of-concept of TITANS neural memory for discrete-item recommendation, with a **structural** forgetting-prevention protocol.
- C3** Comprehensive full-ranking evaluation across 4 benchmarks – accuracy, diversity, and robustness.

CONGA: Three Orthogonal Extensions to SASRec

Phase 1 — train base encoder

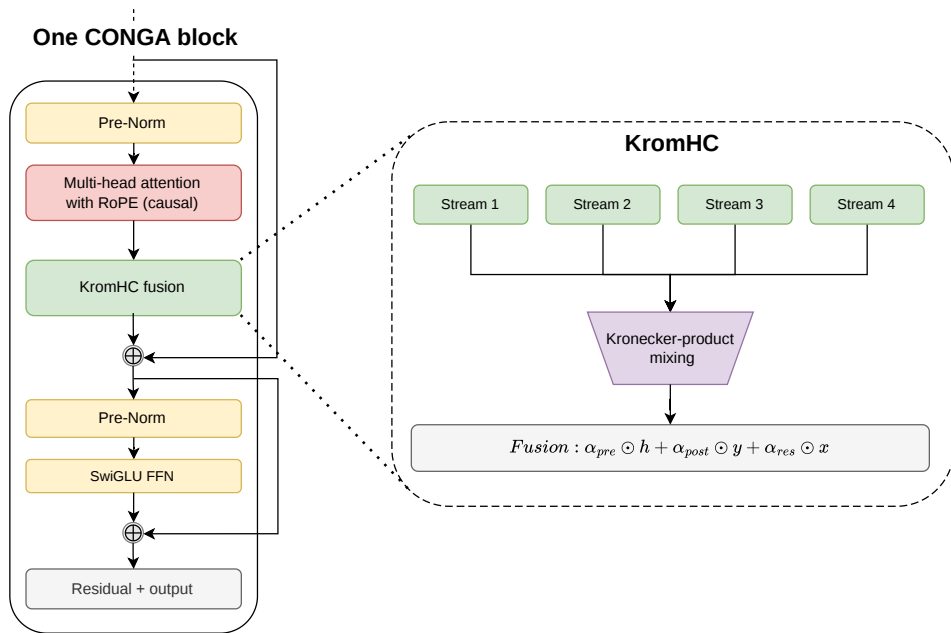


Phase 2 — freeze encoder, train memory



1. **Modernized encoder** – RoPE attention + KromHC multi-stream fusion + SwiGLU FFN
2. **TITANS neural long-term memory** – fixed-size associative memory, added only in Phase 2
3. **Two-phase training protocol** – integrates memory without disturbing the learned encoder

Modernized Encoder: RoPE + KromHC + SwiGLU



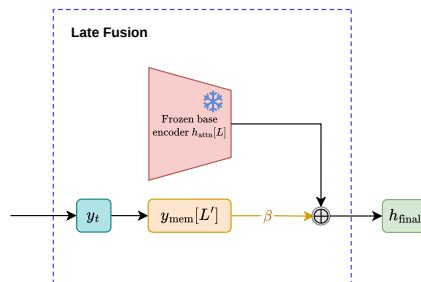
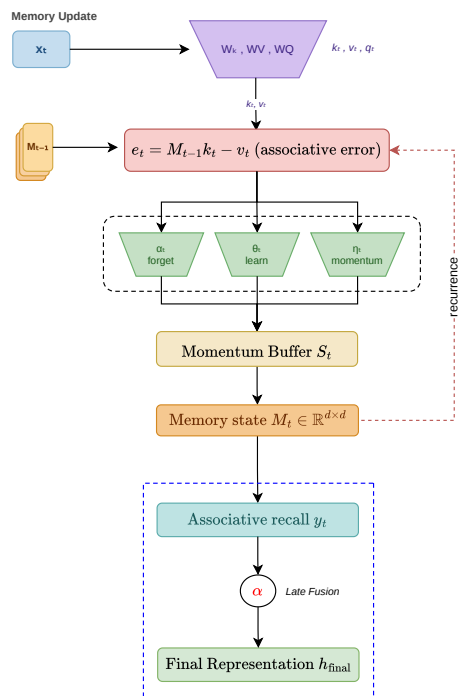
RoPE rotates queries/keys by position; the rotation matrix is orthogonal ($\|\mathbf{R}_{\theta,t}x\| = \|x\|$ for **any** t) so attention stays well-scaled beyond L_{train} (+3.9% NDCG@10 on ML-1M from RoPE alone).

KromHC replaces the single-stream residual with M parallel streams, fused by an **exact** doubly-stochastic Kronecker-product matrix – $O(1)$ from 2 scalars vs. iterative Sinkhorn-Knopp. Adaptive stream count: $M = 4$ if mean history length $\bar{L} \geq 50$, else $M = 2$.

SwiGLU replaces ReLU in the FFN, keeping gradients non-zero everywhere.

TITANS Memory + Two-Phase Training

TITANS Neural Long-Term Memory



TITANS compresses the full history into $M_t \in \mathbb{R}^{d \times d}$, updated online from the associative error $e_t = M_{t-1}k_t - v_t - O(d^2)$ cost, independent of sequence length.

Late fusion at the final position only:

$$h_{\text{final}} = h_{\text{attn}}[L] + \beta \cdot y_{\text{mem}}[L']$$

($L = 300$ attention window, $L' = 600$ extended memory window)

Two-phase protocol

Phase 1: train the encoder (no TITANS). Phase 2: **freeze** the encoder, train only TITANS. Structural guarantee – Phase 2 can only **add** the memory term, never degrade Phase-1 predictions.

Custom CUDA Kernels

Kernel	Role	Fwd	Bwd
RoPE	Fused query/key rotation	5.0×	5.4×
SwiGLU	Fused 3-pass FFN	3.9×	3.5×
KromHC	Closed-form vs. Sinkhorn-Knopp	5.2×	4.8×
TITANS	Register tiling + re-materialization	13.2×	15.0×
Total	All four kernels combined	10.0×	10.9×

- ▶ TITANS kernel: $37.5 \times$ activation-memory reduction $\rightarrow$ Phase 2 at $L' = 600$ fits in 12 GB VRAM
- ▶ Full 300-epoch training run completes in under 5 hours (RTX 5090)

Datasets, Baselines, Protocol

Stat	Beauty	Yelp	Steam	ML-1M
# Users	22,363	30,431	88,310	6,040
# Items	12,101	20,033	3,581	3,416
Avg. seq.	8.9	10.4	14.2	165.6
Sparsity	99.93%	99.95%	99.60%	95.20%

9 baselines: Caser, GRU4Rec, SASRec, BERT4Rec, FMLPRec, DuoRec, FEARec, BSARec, WEARec (AAAI'26)

Protocol: leave-one-out + strict **full-ranking** (all-item scoring, no sampled metrics)

RQ1: CONGA vs. the Strongest Baseline

Dataset	Metric	Best baseline	Value	CONGA	Improv.
Beauty	HR@5	FEARec	0.0706	0.0718	+1.56%
	NDCG@10	FEARec	0.0601	0.0594	-1.16%
Yelp	HR@5	WEARec	0.0492	0.0551	+11.99%
	NDCG@10	WEARec	0.0412	0.0467	+13.35%
ML-1M	HR@5	WEARec	0.2051	0.2435	+18.72%
	NDCG@10	WEARec	0.1694	0.2012	+18.77%
Steam	HR@5	DuoRec	0.0754	0.0779	+3.32%
	NDCG@10	DuoRec	0.0646	0.0672	+4.02%

Key result

CONGA ranks first on **14/16** full metric-dataset combinations; gains scale with interaction density and history length.

RQ2: Incremental Ablation

ID	Config	H@5	N@5	H@10	N@10
Yelp – sparse, $\bar{L} = 10.4$					
A0	Base encoder (RoPE)	0.0554	0.0365	0.0861	0.0463
A1	A0 + KromHC	0.0524 ↓	0.0339 ↓	0.0815 ↓	0.0433 ↓
A2	A1 + SwiGLU	0.0534 ↑	0.0350 ↑	0.0829 ↑	0.0444 ↑
A3	A2 + TITANS (full)	0.0551 ↑	0.0355 ↑	0.0901 ↑	0.0467 ↑
ML-1M – dense, $\bar{L} = 165.6$					
A0	Base encoder (RoPE)	0.2434	0.1675	0.3392	0.1984
A1	A0 + KromHC	0.2425 ↓	0.1669 ↓	0.3411 ↑	0.1988 ↑
A2	A1 + SwiGLU	0.2432 ↓	0.1698 ↑	0.3389 ↓	0.2007 ↑
A3	A2 + TITANS (full)	0.2435 ↑	0.1700 ↑	0.3402 ↑	0.2012 ↑

Contributions are **regime-dependent**: KromHC dips on sparse Yelp, but TITANS gives the largest single gain there (A2→A3: +5.2% NDCG@10). On ML-1M, all three components compound – A3 strictly beats A0 on every metric.

RQ3: Diversity and Robustness (ML-1M, top-10)

Metric	Base	Full CONGA	Δ
Coverage@10	74.68%	74.80%	+0.12 pp
ILD (diversity)	0.6528	0.6235	-0.029
Tail Ratio	0.94%	0.77%	-0.17 pp
Head Bias	5.58%	6.36%	+0.78 pp
Repetition Rate	0.00%	0.00%	± 0

Reading

A small, classical accuracy–diversity trade-off – not a structural bias. Repetition Rate stays **exactly zero**, confirming correct interaction masking.

RQ4: Where Does TITANS Help Most?

Group (history len.)	Users	w/o TITANS	CONGA-full	$\Delta\%$
Short (≤ 50)	1,837	0.2535	0.2536	+0.04%
Medium (51–150)	2,130	0.2024	0.2027	+0.15%
Long (> 150)	2,073	0.1521	0.1532	+0.72%
Overall	6,040	0.2007	0.2012	+0.25%

Relative gain grows **monotonically** with history length – consistent with TITANS' $O(\sqrt{T})$ regret bound. The learned scalar β self-regulates toward zero for short histories, where memory carries little signal.

Regime-Dependent Behavior and Limitations

Practical rule

RoPE generalizes unconditionally. KromHC and TITANS pay off once $\bar{L} \geq 50$; below that, deploy the RoPE-only backbone (A0).

Limitations

- ▶ Beauty/Yelp/Steam are comparatively small, sparse benchmarks; the $\bar{L} = 50$ threshold was chosen qualitatively, not via a dedicated sweep
- ▶ No benefit for cold-start users (< 10 interactions)
- ▶ $O(d^2)$ -per-user memory footprint restricts batch size at large d
- ▶ TITANS is a frozen-encoder **patch**, not a unified self-modifying architecture (cf. Nested Learning / HOPE) – left as future work

Conclusion

CONGA transfers RoPE, KromHC, SwiGLU, and TITANS neural memory into sequential recommendation via a two-phase forgetting-prevention protocol.

Headline numbers

+18.72% HR@5 on ML-1M, +13.35% NDCG@10 on Yelp, zero repetition rate across all configurations, first-place on 14/16 metrics.

Principled transfer of LLM architectural advances to recommender systems is tractable and effective – and points to a broader blueprint for closing the gap between language modeling and recommendation.

Thank you.

Happy to take your questions.

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